

① Heap :- There are 2
types of Heap :-

1. Min Heap

2. Max Heap

① Min Heap :-

Parent

Node's

Value

< =

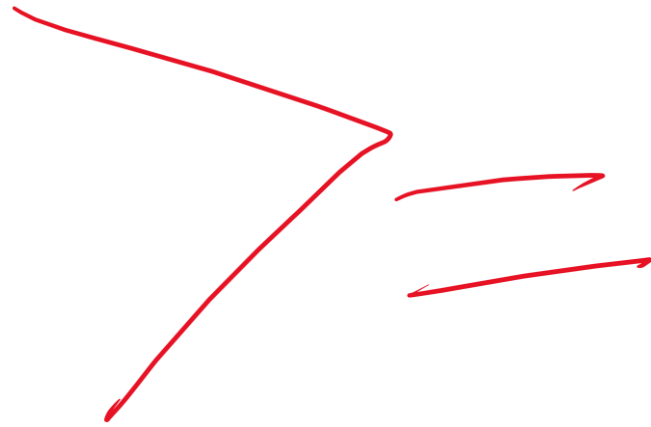
Children

Node's

Value

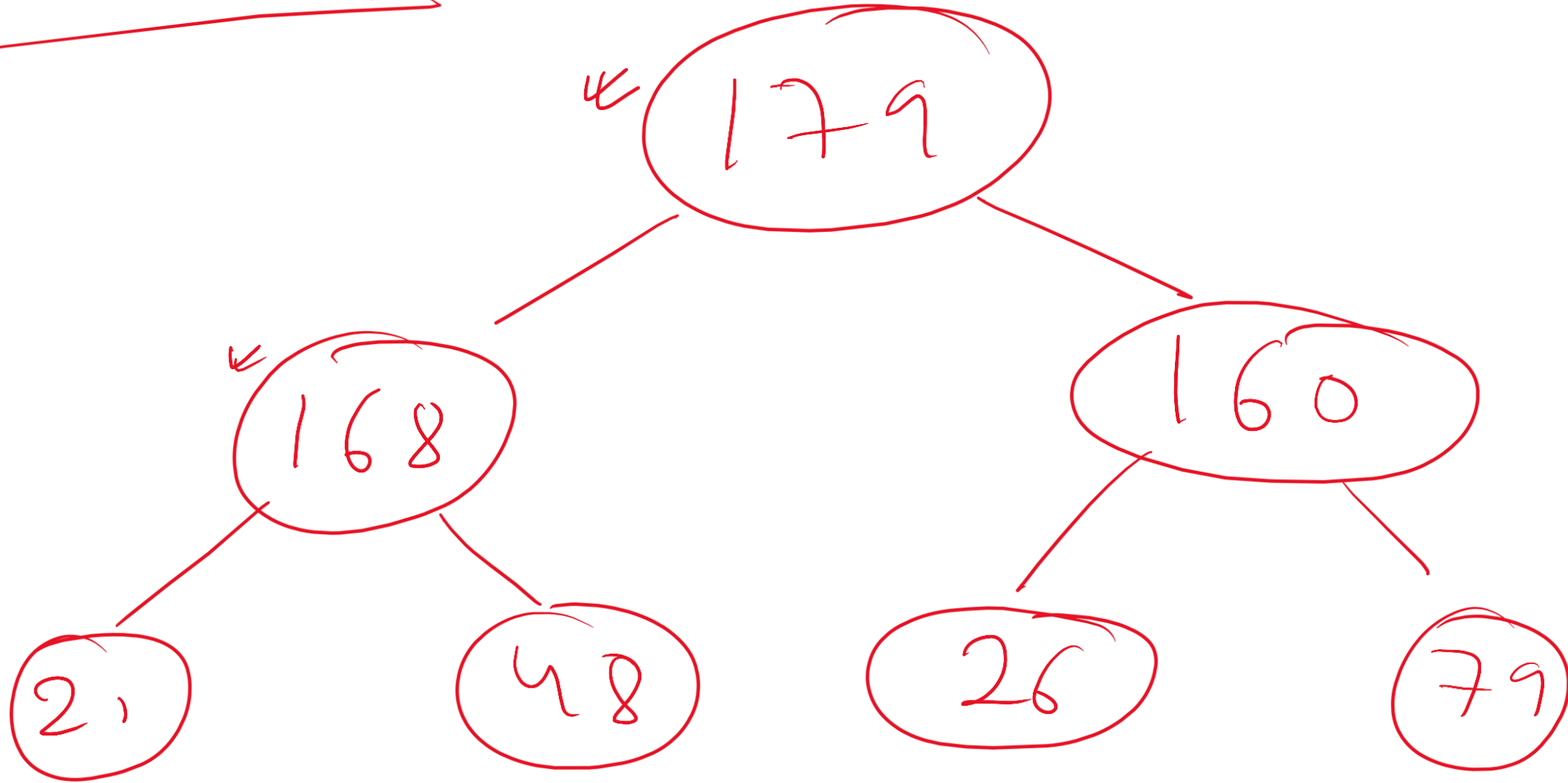
① Max Heap :-

Parent
Node
Value

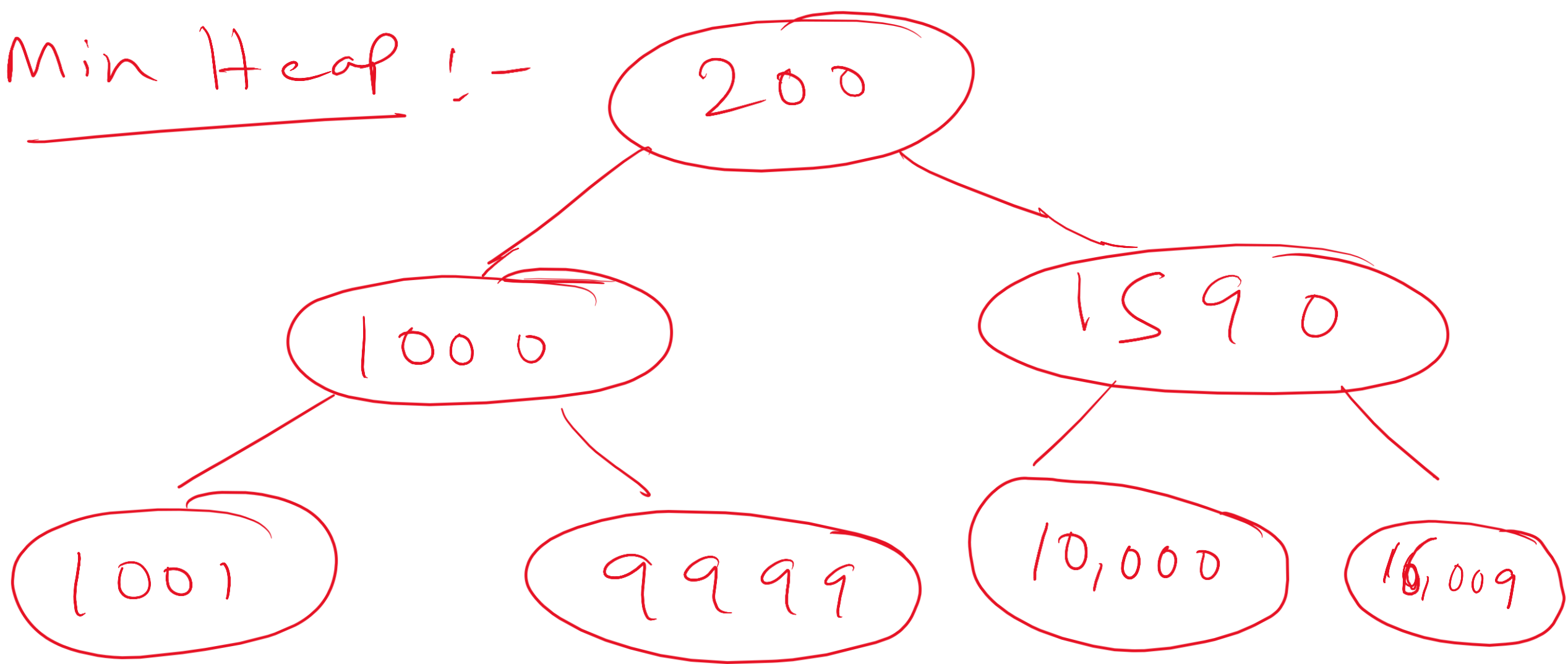


Children
Node
Value

eg:- max Heap:-

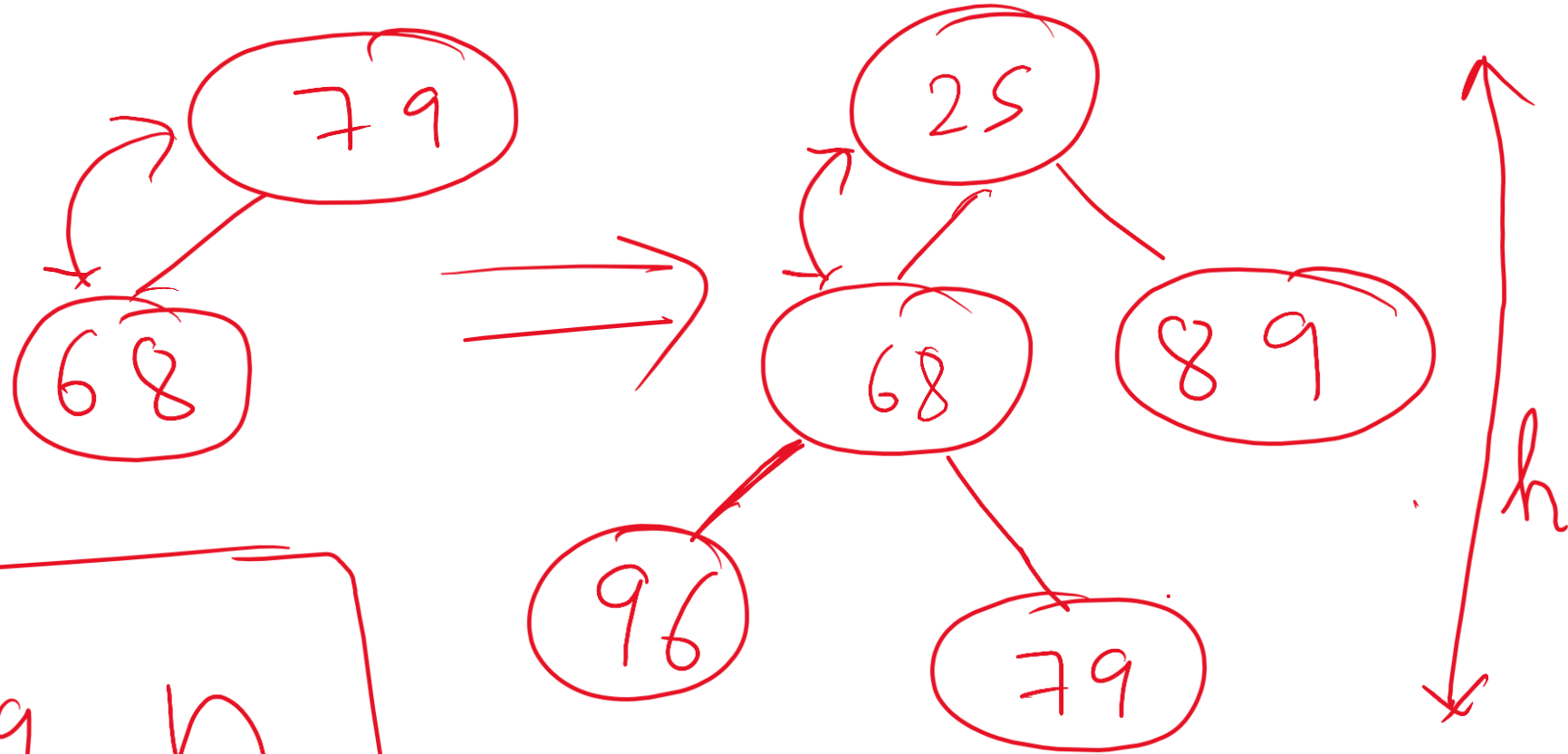


Min Heap :-



① Heapification :- 79, 68, 89, 96, 25

minHeap :-



$$h = \log_2 n$$

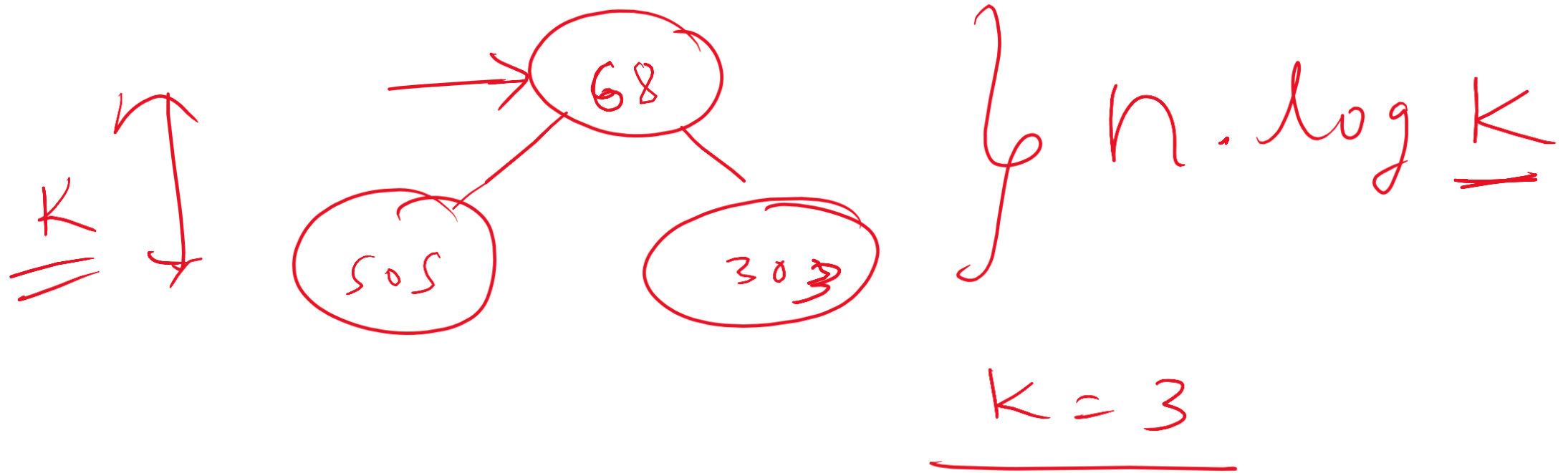
① In Min Heap, minimum element will always be at root Node of the Tree.

① In Max Heap, maximum element will always be at root Node of the Tree.

① Whenever in an Interview, you are
IMP asked to find K^{th} Largest element
in that case use MIN HEAP

and to find K^{th} Smallest
element, use MAX HEAP
of size K .

eg:- 29, 68, 30, 5, 4, 303, 505
 $K=3$, Find K^{th} largest element



① Priority Queue by default implements Min Heap.

② Max Heap :-

PriorityQueue<Integer> maxHeap =
new PriorityQueue<>(Comparator.
reverseOrder());